

Pavan Kumar Nayakanti

Enterprise AI & Cloud Architect | LLM Systems | RAG Pipelines | MLOps | Multi-Cloud | FinTech

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SUMMARY

20+ year enterprise architect and AI platform leader with a career spanning global banking institutions -- ADP, Charles Schwab, BMO, RBC, and Microsoft. Architected production AI systems (GenAI, LLM, RAG-style retrieval pipelines, HITL control layers, and AI observability frameworks) in regulated FinTech environments. Deep expertise in cloud-native architecture across AWS, GCP, and Azure; Kubernetes (CKAD), Terraform, CI/CD governance, and distributed systems at scale. Defined enterprise architecture governance frameworks (CPMI) and reference architectures adopted across three global banking clients. Masters in Artificial Intelligence (2024); GCP Professional Cloud Architect; Azure Solutions Architect Expert; AI Engineer Associate (AI-102) and Data Scientist Associate (DP-100) certified. Combines architectural leadership with hands-on production delivery and C-suite stakeholder management.

KEY ACHIEVEMENTS

**Enterprise LLM Platform (BMO)**
Architected production GenAI platform (Azure OpenAI + AWS Bedrock): RAG-style retrieval pipelines, enterprise guardrail architecture, HITL control layers, compliance audit logging, and AI observability for regulated banking.

**Enterprise Architecture Governance**
Defined Cloud Platform Maturity Index (CPMI) -- architecture governance framework scoring cloud security, automation, and capability maturity across ADP, Charles Schwab, and BMO.

**Cloud-Native Platform Architecture**
Architected modeas (200+ builds/hour, 10,000+ manual hours eliminated) and BMO's HA+DR AWS migration (CDK, ECS Fargate, Lambda) -- production-grade, multi-cloud at banking scale.

**Architecture Team Leadership**
Built Charles Schwab's Platform Engineering team from zero (25 members in one quarter); scaled ADP's SRE org from 4 to 40 engineers; reduced MTTR 40%+ through AIOps-driven remediation.

EXPERIENCE

Sr. Specialist - Cloud Practice & SRE Lead

IRIS Software - India

Sep 2020 - Present | Clients: ADP, Charles Schwab, BMO

- Architected **enterprise GenAI platform on Azure OpenAI and AWS Bedrock for BMO**: designed RAG-style retrieval pipelines (BCRG/LTS), enterprise guardrail architecture, HITL control layers, compliance-grade audit logging, and LLM observability -- production deployment with full regulatory sign-off at a top-5 North American bank.
- Defined **AI observability frameworks** for production LLM workloads: drift detection, response quality monitoring, token usage tracking, latency baselines, and hallucination guardrails for regulated financial data.
- Designed **Cloud Platform Maturity Index (CPMI)** -- enterprise architecture governance framework defining reference architectures, scoring cloud security, automation, and capability maturity across three global banking clients (ADP, Charles Schwab, BMO).
- Executed **multi-cloud architecture strategy**: AWS CDK/ECS Fargate/Lambda production stack for BMO with HA+DR designed in at inception; GCP landing zone architecture and workload migration POCs for Charles Schwab evaluating GCP-native and Kubernetes-based deployments.
- Governed **IaC, CI/CD, and container orchestration** at scale: Terraform-driven infrastructure, Kubernetes/Helm container pipelines, Jenkins/GitHub Actions workflows, self-healing deployment architecture.
- Implemented **AIOps-driven automated remediation** -- Ansible + Python; Prometheus, Grafana, and Dynatrace for full-stack observability, distributed tracing, and root cause analysis; reduced MTTR 40%+.
- Defined **SLOs, SLIs, and error budget governance** across three global banking clients; presented architecture strategies, AI roadmaps, and reliability outcomes directly to VP and C-suite stakeholders.
- Scaled ADP's SRE/Platform organization from **4 to 40 engineers** across India, Canada, and US; built Charles Schwab's 25-member Platform Engineering team from the ground up in one quarter.
- Drove **AWS FinOps architecture for BMO**: cost governance framework, rightsizing, tagging enforcement, cloud waste reduction -- performance optimization and cost discipline at enterprise scale.

Automation Technology Specialist

IRIS Software - Toronto, Canada

Jun 2018 - Sep 2020 | Client: Royal Bank of Canada Insurance

- Established **SRE and DevSecOps architecture** within RBC's engineering programme: observability standards, reliability-focused deployment gates, incident classification frameworks across 250+ applications.
- Architected **SAST, DAST, and IAST security pipeline** -- SonarQube, JFrog Artifactory, Jenkins, and GitHub for shift-left security and dependency governance across Java and multi-stack workloads.
- Designed and delivered **Robo Deployment Framework**: governed CI/CD orchestration platform with RTB/CTB pipeline segregation, SOX-compliant change evidence, and self-service release for 250+ applications across 60+ teams; reduced onboarding from 5 weeks to 5 hours.
- Inducted **DORA framework** as enterprise architecture standard across RBC Wealth Management and Insurance -- Deployment Frequency, Lead Time, Change Failure Rate, MTTR as primary engineering KPIs.
- Migrated **250+ applications across 60+ teams** from CA SCM/Harvest to GitHub Enterprise -- architected phased migration with automated evidence, audit-ready from day one.

Automation Architect

IRIS Software - Toronto, Canada

Feb 2017 - Jun 2018 | Clients: Insurance & Capital Markets

- Designed **container-orchestrated, fault-tolerant deployment pipelines** using Kubernetes and Helm for zero-downtime rollouts across capital markets platforms with Kafka-integrated event-driven workloads.
- Pioneered **Cognitive and Intelligent Process Automation (CPA/IPA)** -- automated 80+ applications and 250+ components across heterogeneous stacks (Java, NodeJS, Kafka, Salesforce, Mainframe).

Technology Manager

TechMahindra Pvt. Ltd. - Noida, India

Jul 2015 - Feb 2017 | Client: Adobe

- Architected zero-downtime enterprise platform migration during Adobe's data centre relocation; led SAP HANA modernisation and end-to-end data pipeline re-engineering at enterprise scale.

Sr. SDET - QA, Automation & Production Engineering

Microsoft Corporation - India

Jul 2007 - Dec 2014 | Sales & Marketing Platforms (US, EMEA, APAC)

- Owned full quality and production engineering lifecycle for global Sales & Marketing platforms; defined and enforced non-functional architecture standards across network, infrastructure, and application tiers -- sustained **99.99% uptime for 7 years** across US, EMEA, and APAC.
- Architected and delivered QA automation frameworks integrated with CI/CD; managed production incident lifecycle, RCA, and infrastructure engineering across Windows and Linux in an Agile environment.

Senior Software Testing Engineer

Semantic Space Technologies - India

Jan 2005 - Jun 2007

- L3 Analyst and release engineering lead for Statusmart application (3,000+ sales users, Bank of America); owned ETL development, SQL optimisation, and incident management.

Software Consultant

Defense Electronics Research Laboratory (DRDO) - India

Oct 2002 - Dec 2004

- Systems development and architecture consulting for defence-grade applications requiring mission-critical reliability and security standards.

EDUCATION

Masters in Artificial Intelligence Woolf University	2024
General Management Program (GMP) ISB Hyderabad	2022
M.Sc in Computer Science Osmania University	2002

CERTIFICATIONS

Microsoft Certified: Azure AI Engineer Associate (AI-102)

Microsoft

Microsoft Certified: Azure Data Scientist Associate (DP-100)

Microsoft

Certified Kubernetes Application Developer (CKAD)

CNCF - 2025

Google Cloud Certified - Professional Cloud Architect (PCA)

Google Cloud - 2026

Microsoft Certified: Azure Solutions Architect Expert (AZ-305)

Microsoft

AWS Certified Solutions Architect - Associate (SAA-C01)

Amazon Web Services - 2026